

Gene Sight Eva-Female Infertility NGS Panel

Patient name	:	Mrs. XX	PIN	:	XXX
Gender/Age	:	Female / XX Years	Sample number	:	XX
Referring Clinician	:	XX	Sample collection date	:	NA
Hospital/ Clinic	:	XX	Sample received date	:	NA
Specimen		Peripheral blood	Report date	:	NA

INDICATION FOR TESTING

Proband, Mrs. XX is of consanguineous marriage. Her AMH is 0.86 and she has undergone 2 ART cycles with self-gametes. The first cycle yielded 08 oocytes, and 03 blastocysts were formed. 02 blastocysts were transferred, and the result was negative. The second cycle yielded 05 oocytes (poor quality), and no embryos were formed. Proband, Mrs. XX have been evaluated for pathogenic variations and polymorphisms.

RESULTS

FEMALE FERTILITY SINGLE NUCLEOTIDE VARIATION ANALYSIS

Likely Pathogenic Variant related to the given phenotype was identified

List of significant variant identified related to the phenotype:

Gene	Region	Variant*	Allele Status	Disease	Classification*	Inheritance Pattern
PATL2 (-)	Exon 9	c.604C>T (p.Gln202Ter)	Homozygous	Oocyte/zygote/embryo o maturation arrest 4 (OMIM#617743)	Likely Pathogenic	Autosomal recessive

*Genetic test results are based on the recommendation of American college of Medical Genetics [1-5].
No other variant that warrants to be reported for the given clinical indication was identified.

FEMALE POLYMORPHISM ANALYSIS

List of polymorphic variants (SNP) identified:

Gene Name	RsID	Variant	Genotype	Zygosity	Observed Phenotype	Recommendations
AMH	rs10407022	c.146G>T	TT	Homozygous Polymorphism	Associated with reduced AMH bioactivity and altered follicular regulation. Lower functional ovarian reserve and reduced responsiveness to mild ovarian stimulation. [4-7].	Avoid minimal or low-dose stimulation protocols. Increased FSH dosing and extended stimulation protocols. Adjuvant therapies (DHEA, growth hormone, CoQ10) supplementation.
IL10	rs1800872	c.-627A>C	CC	Homozygous Polymorphism	Immunological susceptibility factor in implantation failure or recurrent pregnancy loss [8-10].	Peripheral & endometrial NK cell assessment. Th1/Th2 cytokine profile (functional testing)
KISS1	rs4889	c.242C>G	GG	Homozygous Polymorphism	KISS1-mediated maturation defects are associated with suboptimal endogenous LH surge — exogenous hCG trigger bypasses this, but cytoplasmic maturation deficits may persist. [11-13]	Dual trigger can be considered.
MTRR	rs1801394	c.66A>G	AG	Heterozygous Polymorphism	Partial reduction of MTRR enzymatic activity. Suboptimal embryo competence. Susceptibility for Recurrent implantation failure (RIF). Early pregnancy loss / recurrent miscarriage (RPL) [14-15].	Evaluate serum homocysteine, Vitamin B12 levels. Vitamin B12 supplementation.

Variant Interpretation

Interpretation of significant variant identified in Mrs. XX related to the phenotype

PATL2: c.604C>T

Variant summary: A homozygous stop gained variation in exon 9 of the *PATL2* gene (ch15:g.44672068G>A, NM_001387263.1, Depth: 147x) that results in the premature truncation of the protein at codon 202 (p.Gln202Ter) was detected. This variant is a stop gained variant, which occurs in an exon of *PATL2* upstream where nonsense-mediated decay is predicted to occur. This variant is predicted to cause loss of normal protein function through protein truncation.

Population frequency: This variant has minor allele frequency of 0.0009% in gnomAD database and has minor allele frequency of 0.02% in 1000 genomes database.

In-silico predictions: The *in-silico* predictions of the variant are deleterious CADD and MutationTaster2. The reference codon is conserved across mammals in PhyloP and GERP++ tools.

OMIM phenotype: Oocyte/zygote/embryo maturation arrest 4 (OMIM#617743) is caused by homozygous or compound heterozygous mutation in the *PATL2* gene (OMIM*614661). Oocyte maturation defects due to mutation in *PATL2* show phenotypic variability, with some oocytes exhibiting maturation arrest at the germinal vesicle stage and others at the metaphase I stage. In some patients, a few oocytes progress to polar body I; those oocytes either undergo fertilization failure or, in those that are fertilized, early embryonic arrest. This disease follows Autosomal recessive pattern of inheritance [2].

Variant classification: Based on the evidence, this variant is classified as likely pathogenic. **In this view, clinical correlation and familial segregation analysis are strongly recommended to establish the significance of this finding. If the results do not correlate, additional testing may be considered based on the phenotype observed.**

Recommendations

- Sequencing the variant(s) in the partner and other affected and unaffected members of the family is recommended to confirm the significance.
- Genetic counseling is recommended.

Methodology: Single Nucleotide Variation - Single Nucleotide Polymorphism

SNV analysis: DNA extracted from the blood, was used to perform targeted gene capture using a custom capture kit (for SNV analysis – 845 genes). The targeted libraries were sequenced to a targeted depth of 80 to 100X using SurfSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions

including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (V4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. *In-silico* predictions of the variant were carried out using VS-SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

SNP analysis: Variant analysis (for SNP analysis – 30 genes) and interpretation is done using VarSeq Software. Extensive scientific literature, Information from variant analysis and disease specific databases, population specific research is used to interpret and recommend. All results are finally approved by medical geneticists.

Sequence data attributes

Total reads generated	13.62 Gb
Data ≥ Q30	96.46 %

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.

Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.
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Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR, Promoter region variants and CNV are not assessed using this assay.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- In addition, this test does not cover detection of such structural variants in the genes *SMN1*, *SMN2*, *HBA1*, *HBA2*, *PMS2*, *GBA* and *CYP21A2* which have very high homology. Moreover, the ORF15 transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].
- Result interpretation was done based on the literature evidence available at the time of reporting. The clinical significance of the polymorphic variants tested can change over time and Anderson Diagnostics & Labs cannot be held responsible for this.
- This is not a diagnostic test and so not to be considered as diagnosis of any disease. This test is meant only for understanding the polymorphism at a given position and its association with various clinical parameters.

References

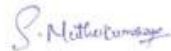
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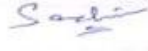
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